Project - Low Level Design
On
Project Title

Real Estate Support Triage Agent

Course Name: Agentic AI

Institution Name: Medicaps University – Datagami Skill Based Course

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Project Number:GAI-30

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Academic Year: 2025-26

1. Introduction

The Low Level Design (LLD) describes the internal working of the Real Estate Support Triage Agent, an intelligent system designed to automate the processing of customer communications such as inquiries, complaints, and service requests.
The system leverages:
Natural Language Processing (NLP)
Named Entity Recognition (NER)
Large Language Models (LLMs)
It ensures:
Efficient message classification
Automated prioritization
Information extraction
Smart response generation
The system is modular to ensure scalability, maintainability, and real-time processing.

1.1 User Interface Module (Dashboard / API Layer)

Technology Used: Python (FastAPI) or Web UI (React)
Responsibilities:

- Accept incoming customer messages (email/chat/API)
- Display classified results (intent + urgency)
- Show extracted entities
- Display generated response drafts

Inputs:

- Customer message (text/email/chat)
- Optional metadata (timestamp, user ID)

Outputs:

- Classified intent (Inquiry / Complaint / Request)
- Urgency level (High / Medium / Low)
- Extracted entities
- AI-generated response draft

1.2 Input Validation Module

Purpose:
Ensures incoming messages are valid and preprocessed before analysis.

Validations Performed

- Empty message check
- Language detection

- Text cleaning (remove noise, special characters)
- Length validation

Outcome

- Cleaned and normalized text
- Validation status (True/False)
- Error message (if invalid)

1.3 Text Classification Module

Classifies:

Intent → Inquiry / Complaint / Support Request
Urgency → High / Medium / Low

Techniques Used

Pre-trained NLP models
Rule-based keyword boosting (for urgency)

Core Methods

classify_intent(message)
detect_urgency(message)
priority_score(message)

1.4 Entity Extraction Module

Finds important details from the message.

Extracted Data

- Property ID
- Date
- Location
- Customer name
- Booking details

Example
Message: "I want to visit property P123 on 20 Feb"
→ Property ID = P123
→ Date = 20 Feb

1.5 Response Generation Module (LLM Integration)

Purpose:

Generates context-aware response drafts.

Responsibilities

Construct prompt using:

- Intent
- Urgency
- Extracted entities

Call LLM API
Format response professionally

Core Methods

- generate_prompt(message, intent, entities)
- generate_response(prompt)
- format_response(response)

1.6 API Integration Module

Purpose
Handles secure communication with LLM APIs.

Responsibilities

- API key management (env variables)
- Send HTTP requests
- Handle failures & retries
- Manage rate limits

Security Measures

- No hardcoded API keys
- HTTPS communication
- Exception handling

1.7 Workflow Orchestration Module

Purpose:
Coordinates all modules and ensures smooth pipeline execution.

Responsibilities

- Manage message flow
- Trigger modules in sequence
- Handle errors

2. Class Design

Class: UserInterface

Methods:

- receive_message()
- display_results(data)
- show_error(msg)

Class: InputProcessor
Methods:
- clean_text(message)
- validate(message)

Class: Classifier
Attributes:
- model_name

Methods:
- classify_intent(message)
- detect_urgency(message)

Class: NERExtractor
Methods:
- extract_entities(message)
- format_entities(data)

Class: ResponseGenerator
Attributes:
- api_key
- model_name

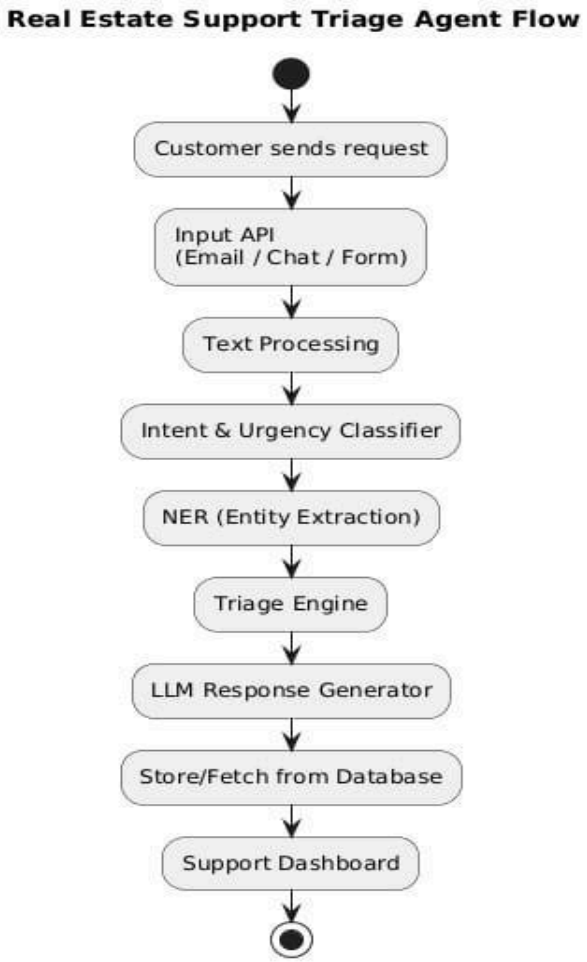
Methods:
- generate_prompt(data)
- call_api(prompt)
- format_output(response)

Class: Orchestrator
Methods:
- process_message(message)
- route_priority(data)

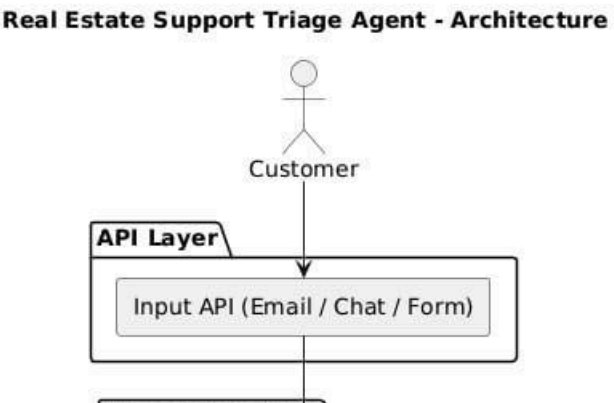
3. Detailed Process Flow

- Customer sends message (email/chat/API)
- UserInterface receives input
- Input-processor validates and cleans message

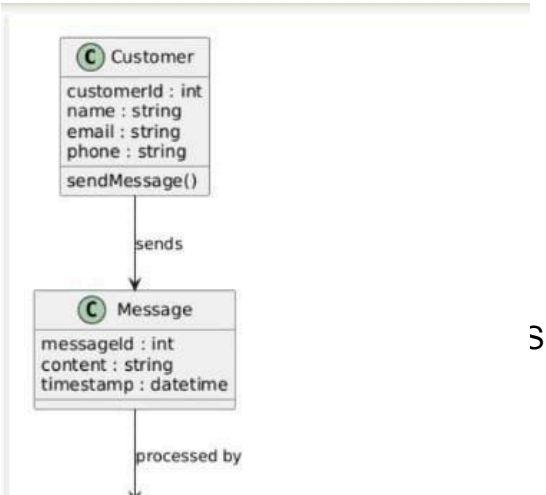
- If validation fails → error displayed
- If valid → Classifier determines intent & urgency
- NER module extracts key entities
- Orchestrator assigns priority
- ResponseGenerator constructs



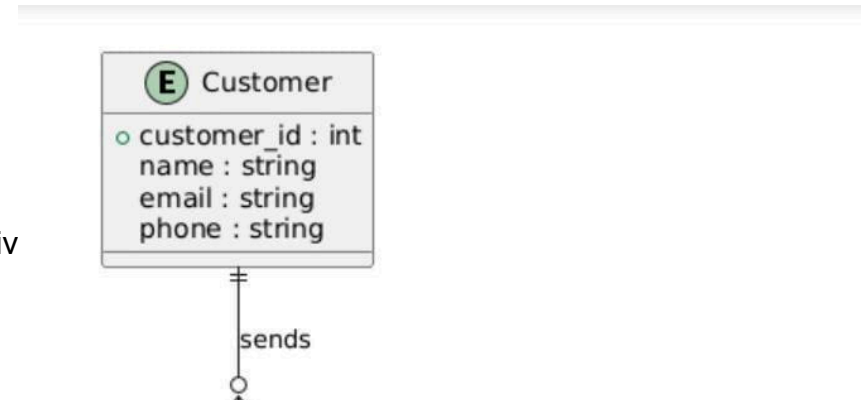
Flow Diagram



Architecture Diagram



Class Diagram



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